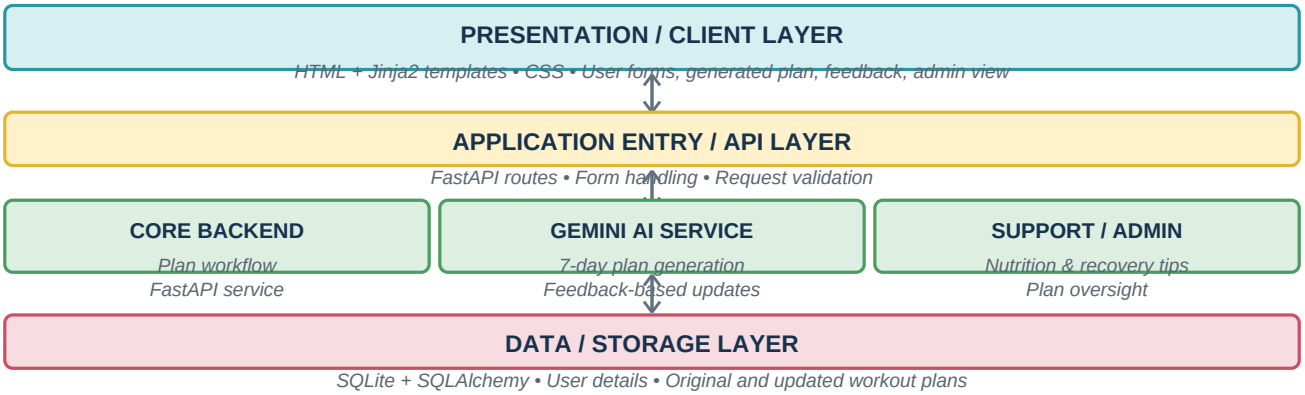
SOLUTION ARCHITECTURE

Date	01 October 2026	**Maximum Marks**	5 Marks
Team ID	SWTID-2026-2430	**Project Name**	FitBuddy – AI Fitness Plan Generator using Gemini Models

Solution Architecture Diagram

High-level view of the FitBuddy web application, backend workflow, Gemini integration, and persistent storage.



External integration: Google Gemini models (API access required)

Data flow: User enters fitness details → FastAPI processes the request → Gemini generates a structured 7-day plan and practical tip → the result is displayed → feedback can trigger an updated plan → SQLite stores user details and plan history.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects user goals and details; displays the generated plan, nutrition/recovery tip, feedback form, and admin view.	HTML, CSS, Jinja2
API / Application Layer	Receives form submissions, validates requests, routes plan-generation and feedback actions, and connects frontend to backend logic.	FastAPI, Uvicorn
Gemini AI Service	Creates a personalized 7-day workout plan and revises it using user feedback; supports structured prompt/output handling.	Google Gemini models
Data / Storage Layer	Stores user details and original/updated workout plans so plan history can be retained and reviewed.	SQLite, SQLAlchemy

Team Members and Work Allocation

Responsibilities are assigned in the same order as the provided team-member list.

S. No.	Team Member	Assigned Responsibility	Expected Contribution
1	Divya.D	Project coordination & requirements	Define target users, fitness goals and inputs; coordinate tasks and consolidate documentation.
2	Deepika.M	Gemini AI integration & prompt design	Create prompts for structured 7-day plans and feedback-based plan updates using Gemini models.
3	Dharani.T	FastAPI backend & routing	Handle form processing and routes for plan generation and feedback; connect routes to AI functions.
4	Divya shree.R	Frontend & user experience	Build HTML/Jinja2 pages for inputs, results, tips and feedback; support responsive styling.
5	DivyaDharshini.V	Database, testing & deployment support	Support SQLite/SQLAlchemy storage, test main flows, and verify local deployment/API docs.

Shared responsibility: All members review the final output, check that generated plans are clearly presented, and prepare the project demonstration. Fitness suggestions are general information and should be adapted to individual needs.